4–21 Find v_O in terms of v_S in Figure P4–21.

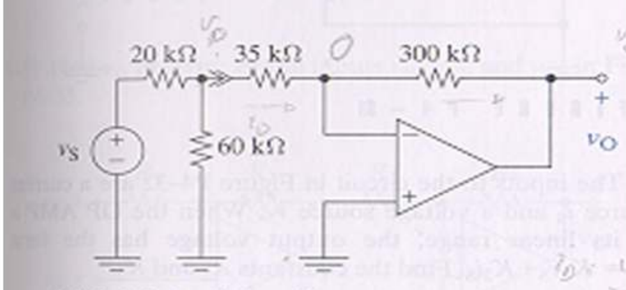

FIGURE P4 – 21

4–24 (a) Find v_O in terms of v_S in Figure P4–24.
(b) Find i_O for $v_S = 2$ V.

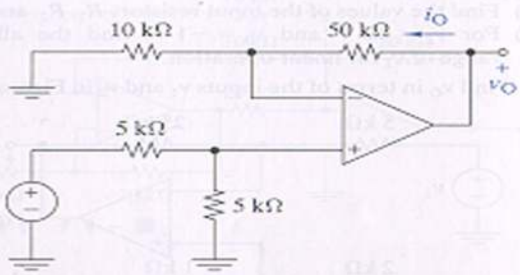

FIGURE P4 – 24

4–28 Find v_O in terms of the inputs v_1 and v_2 in Figure P4–28.

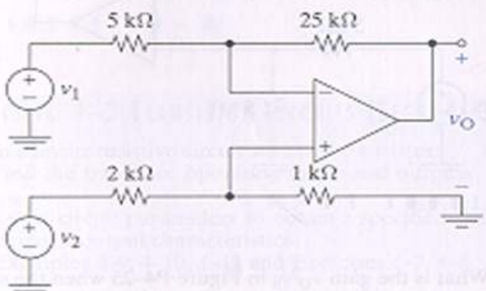

FIGURE P4 – 28

4–29 Find v_O in terms of the inputs v_{S1} and v_{S2} in Figure P4–29.

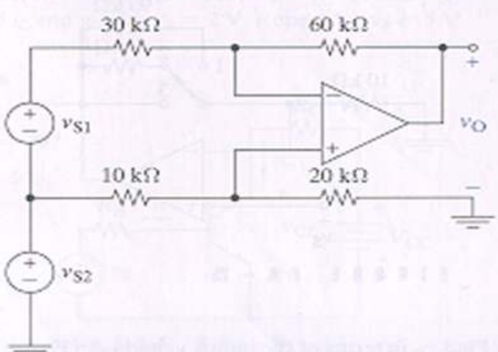

FIGURE P4 – 29

4–31 E It is claimed that $v_O = v_S$ when the switch is closed in Figure P4–31 and that $v_O = -v_S$ when the switch is open. Prove or disprove this claim.

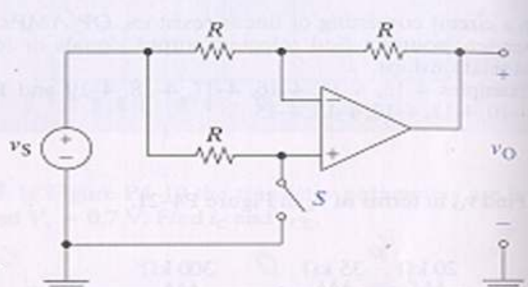

FIGURE P4 – 31

4–32 The inputs to the circuit in Figure P4–32 are a current source i_S and a voltage source v_S. When the OP AMP is in its linear range, the output voltage has the form $v_O = K_1 v_S + K_2 i_S$. Find the constants K_1 and K_2.

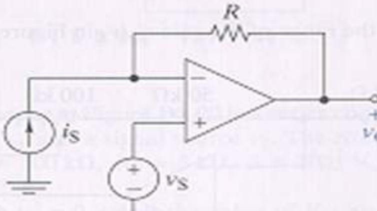

FIGURE P4 – 32

4–35 Find v_O in terms of the inputs v_{S1}, v_{S2}, and v_{S3}, in Figure P4–35.

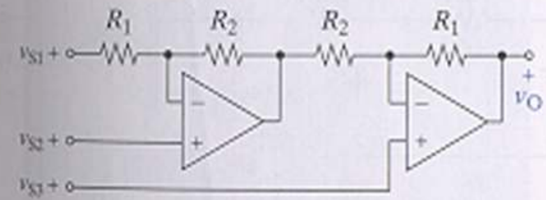

FIGURE P4 – 35

4–38 Find the output v_2 in terms of the input v_1 in Figure P4–38.

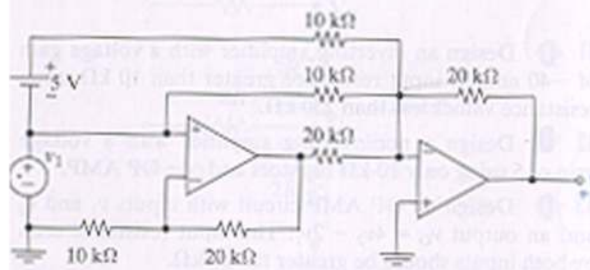

FIGURE P4 – 38

4–45 D Using no more than two OP AMPs, design an OP AMP circuit with inputs v_1, v_2, and v_3 and an output $v_O = 3v_1 - 2v_2 - 5v_3$.

4–54 D The output voltage of a semiconductor temperature sensor is $v_S = (10T + 500)$ mV, where T is the temperature in °C. The sensor is to be used to measure temperatures in the range −40°C to 110°C. Design an interface circuit that translates this temperature range to a 0-V to 3-V output voltage range. Use a standard 5-V reference source to obtain any required bias voltage.